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Working Group XII - Interstellar Medium

Contents

Summary

This working group will analyze the interstellar medium (ISM) of AGORA galaxies. This section provides a condensed summary of our preparations, plans, goals and all major progress.

Guiding topics

- *The density and temperature structure of the ISM:*
  - What are the profiles and extents of gas disks?
  - How clumpy is the ISM? What is the relationship between the gas power spectrum and feedback?
  - How does the density/temperature structure compare to observations (HI holes, etc)?
  - How does the ISM density and temperature structure depend on hydro method and feedback prescription?
- *Star formation and stellar feedback in galaxies:*
  - How is feedback reflected in the global and resolved Kennicutt-Schmidt relation compared to the input (subgrid) star formation efficiency?
  - Is feedback able to appropriately disperse simulated GMCs? How does this affect clump migration at low redshift [avoid overlap with the high-redshift galaxies group].
  - How is this affected by resolution, gas fraction, disk mass, redshift, hydro method and feedback prescription?
- *The gas disk in a cosmological context:*
  - What is the gas fraction in galaxies and how does it evolve with redshift/mass?
  - How does accretion affect the ISM? Do cold streams reach the disk (how does this vary with mass/redshift/code)? How does gas velocity dispersion/thickness vary with redshift and why?
  - How does a metallicity-dependent star formation prescription affect the initiation of star formation in galaxies and disk structure?
- *Other ISM properties:*
  - What are the gas-phase metallicities and metallicity gradients of the galaxies? How do they vary with redshift and disk mass?
  - Morphology: spiral structure and appearance of the gas (esp. in dwarfs and at high redshift).

Progress and Results

Code-by-code landmarks

* All participating codes except for Arepo and Nyx are explicitly represented by this group.

Group leadership

Sam Leitner (sleitner@astro.umd.edu) is currently leading this working group. Please contact me if you would like to lead a spin-off project, or co-lead in some capacity.

Working Group XII - Blog

Untitled Post This is the first blog post for the Working Group XII - Interstellar Medium.
Posted Oct 4, 2012, 12:58 AM by Ji-hoon Kim

Showing posts **1 - 1** of **1**. **View more »**

This section is for internal use. Below, group members should share their ideas, help plan goals, note landmark achievements and, ultimately, discuss and post results (note that space is limited so please link plots). At the bottom of the page there is a link to the google group forum.

Group XII contact people by code

Code	Cosmological zoom-in simulator	Code contacts in group XII
ART-I	Daniel Ceverino	Daniel Ceverino
ART-II	Sam Leitner	Nick Gnedin, Andrey Kravtsov, Sam Leitner, Robert Feldmann
ENZO	John Wise	Nathan Goldbaum, Cameron Hummels, Ji-hoon Kim, Brian O'Shea
GADGET	Amit Kashi, Dusan Keres, Justin Read	Amit Kashi, Dusan Keres, Ken Nagamine, Brant Robertson, Emilio Romano-Diaz, Robert Thompson
GADGET-SPHS	Justin Read	Justin Read, Alexander Hobbs
GASOLINE	Sijing Shen	Charlotte Christensen, Piero Madau, Lucio Mayer, Sijing Shen, James Wadsley
RAMSES	Oscar Agertz, Romain Tesyssier	Oscar Agertz, Matteo Tomassetti

Priority questions

Priority should be given to topics that AGORA is uniquely suited to address as a code comparison project. We want to note where the simulation community agrees, as well as how/where code predictions are not robust. Beyond this we want to determine why those differences exist -- whether they are caused by hydro-method, feedback implementation, etc. The result will provide a context for the community to evaluate the results of past and future simulations. Based on this philosophy some high priority directions follow below.

Numerics and implementation

- Comparing the structure of the ISM simulated using different hydro methods: Several successful SPH simulations (Governato, Guedes, Christensen, and Stinson, etc) have shown that the distribution of star formation and thus feedback energy (modulated through a density threshold) is central to regulating the stellar mass of galaxies. It is therefore important to know whether different techniques can converge on the structure (power spectrum, cold gas clumping, cold gas fraction etc) of the ISM given similar subgrid prescriptions.
- Commonalities of feedback effects on the ISM: What common ISM properties result from different implementations of feedback? Do similar implementations produce similar ISM properties or do the details matter? Is this true for both momentum-driven and thermal methods? Within a hydro category (AMR/SPH)?
- To be added...

Science

- Comparison with isolated galaxies: Can the ISM properties of isolated galaxies (Group V) be simply mapped onto the ISM properties of

cosmological galaxies (of similar fgas)? If not, what causes differences: what role does accretion play in gas disk properties (e.g. driving turbulence)?
- To be added...

Simulation agenda

This agenda is a list of runs that we see as critical to answering the scientific and numerical questions listed above. "Priority" simulations should be completed by all participating codes. "Optional" simulations will be run by at least a subset of codes, but are not mandatory because of the additional implementation effort they require.

This agenda is very tentative: the final priorities will depend on what is decided by the Project as a whole.

Priority

- Vanilla run of the 1e12Q IC:
  - Low resolution basic run so that analysis pipeline can be constructed and initial bugs can be worked out.
  - High resolution basic run for comparison of hydro-solver's effect in a tightly controlled, uniform setting.
- "Realistic" run of 1e12Q IC:
  - favored "realistic" flavor of stellar feedback, preferably with some overlap between codes, and with an effort made to agree on kinetic/thermal effect magnitudes.
- To be added...

Optional

- Specific momentum driven stellar feedback model.
- Specific cooling suppression (blast wave like) model.
- To be added...

Potential directions for analysis

We hope to have an unprecedented parameter space available for analysis. The following may be helpful for brainstorming and prioritizing interesting directions (e.g., pick one item from the parameter space list and one from ISM the properties list).

Parameter space dimensions:
Subgrid physics (cooling, star formation, feedback, pressure floor)
Resolution
Cosmological vs isolated galaxies
Halo mass
Halo formation history
ISM setting (i.e. redshift/accretion rate/gas fraction)
Code and hydro method (SPH, AMR, AREPO)

Predicted and observationally accessible ISM properties:
Structure:
thermodynamic state,
disk extents, profiles, power spectra, clumping, gas fractions,
cloud and hole morphologies,
metal content
the star formation - gas connection (the resolved and global Kennicutt-Schmidt relation),
DLA column density distributions.

Dynamics:
velocity dispersions,

galactic fountains (-> outflow group?),
angular momentum evolution,
evolution during mergers.

Progress and results

Discussion forum

https://groups.google.com/forum/#!forum/agora_ism_group12

Google Group XII